# Creating a system of measurements on a cellphone speaker at Dirac Research

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**Abstract**

Dirac Research are interested in increasing the power capacity of a small cellular-phone speaker. By increasing the power, non-linear effects appear, which have to be removed by non-linear compensation algorithms. To do these algorithm some information about the speaker is needed. In this project, a measurement setup for small cellular phone speakers are created to measure the non-linear effects that appear. A laser is used to measure the membrane displacement from the equilibrium. The current and the voltage are measured with an audio amplifier. Finally, the sound pressure is measured with a microphone and an analog to digital converter. All this data is sent to a master micro-controller, which packages and sends the data to a computer, where the data is displayed in MATLAB. The data from the laser-measurement device was created before this project started, but had a problem with unwanted noise in form of measurement-spikes. These spikes were removed, and the variance of the noise was decreased with a factor 5 by using a more robust setup for the laser. The other three measurements were created and are sent through I2S-communication with a 16-bit resolution with no data losses. Some non-linear effects were measured, where a sinus-wave input got a higher total harmonic distortion at higher power.

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Thomas Olofsson

*Examinator:*
Ping Wu

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# 1 Acknowledgement

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Finally, we want to thank Ping Wu, who held this course and has given us much time, effort and encouragement.

# 2 Introduction

Dirac Research develops software algorithm to improve the sound experience of speaker-systems. There is a market for increasing the sound-volume of the miniature speakers inside cellphones but there is some problems that need to be solved first. The two main problems is non-linear effects and heating in the miniature speaker. Dirac has developed non-linear compensation algorithms to compensate for problems but need a very accurate setup to improve and validate the their algorithms. This report will deal with the construction of the measurement system of membrane-displacement, current, voltage and sound-pressure. A setup of the membrane-displacement measurement was already constructed but had some unwanted noise and problems that needed to be removed. A picture of a similar speaker to the one that is used, is shown below in figure 1.

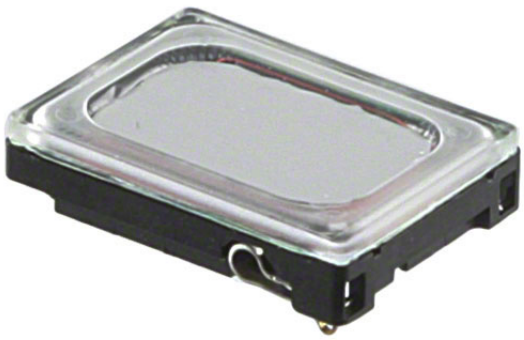

Figure 1: A picture of a miniature speaker similar to the on that is used in this project.[1]

# 3 Theory

## 3.1 A description of a standard loudspeaker

A magnetic circuit is created with two magnets and a metallic core that will lead the magnetic field. Two small gaps are created were two small coils are placed. When a current goes through the coils, a magnetic field is created, which will interact with the magnetic field from the magnets. The coils will then move up or down depending on the current. The movement of the coils will stretch or contract the spider, which will move the membrane and create sound waves.

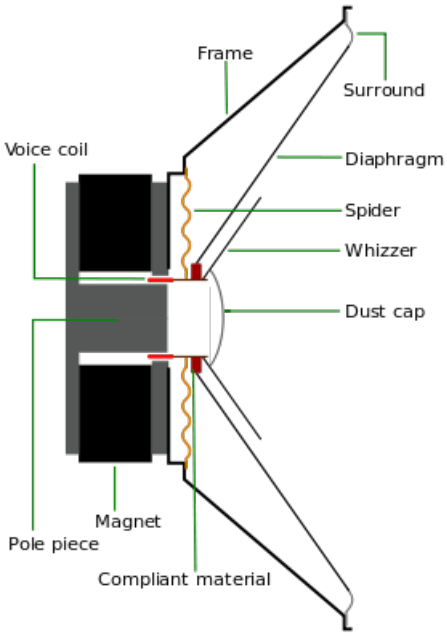


Figure 2: A picture of how a standard loudspeaker are built. Taken from Wikipedia.

### 3.2 Non-linear effects on a speaker

The speaker, in general, behaves linearly whenever the displacement of the coils are small. However, when the displacement is large enough, a number of non-linear effects appears.

Stiffness in the suspension is dependent on the position of the coil. This leads to non-linearity, which will show effect at extreme positions. The force from the displacement of the membrane can be described by Hooks law, where the constant K in this case is a function of the displacement x.

$$F = K(x) * x \tag{1}$$

The stiffness factor $K(x)$ can be written as an polynomial. [2]

$$K(x) = K_0 + \sum_{i=1}^{n} K_i x^i \tag{2}$$

Where $K_i$ are unknown stiffness parameters. This formula shows that when x is small, K(x) will be close to $K_0$ and the system will act linear. When x gets bigger, the coefficients $K_i$ will have a larger effect and the system becomes non-linear.

The impedance of the coil is dependent of the position and the current. The position of the membrane will effect the inductance because of the geometry of the speaker. The impedance will be lower when the coil is above the gap. When the coil is below the gap, it is surrounded by metal giving the magnetic field created by the coil a smaller magnetic resistance. The impedance will also change depending on the current, because the moving coil will induce eddy currents in the iron, which will lead to additional losses. [3]

Force factor, Bl(x) is how strong the speaker-motor is, measured in Tm. Dependent on the geometry of the speaker, i.e. how the magnetic field from the magnet behaves. When the coil is partly removed from the gap where the magnetic field is, the magnetic force factor will decrease. The Force factor is close to constant as long as the coil is close to its equilibrium point. A curve of the force plotted against the displacement is shown in figure 3. If the coil is longer than the gaps width, an overhang is created and the linear part of the force-factor is longer.

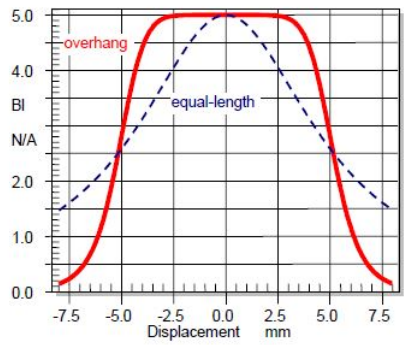


Figure 3: How the force-factor, Bl(x), changes with the displacement of the coil.

The force factor is also dependent on the movement-speed of the coil. A fast moving coil with a current through a static magnetic field will lead to a back-EMF. [4]

### 3.3 Spectral estimate

To find non-linearities in the signal, calculating the spectral estimate can be useful. By checking if the amplitude of the harmonics are bigger, a conclusion can be made that the signal changes properties more than phase and amplitude. One of the easier ways to estimate the spectra of a discrete signal is to use a periodogram with a rectangular window function. The periodogram is basically the absolute value of the fast fourier transform of the signal squared, normalized with the number of samples used. [5]

$$\hat{\Phi}(\omega) = \frac{1}{N}|\sum_{t=1}^{N} y(t)e^{-i\omega t}|^2 \tag{3}$$

$\hat{\Phi}(\omega)$ is the spectral estimate, N is the number of samples and y(t) is the signal.

### 3.4 Scaling

All measurements used two's compliment to represent negative numbers. In MATLAB the numbers received are not in bit-form, they are whole numbers.A formula to convert from a hole number to the corresponding negative number is needed. Suppose a whole number y is received. If $y \geq 2^{N-1}$, then y is a negative number. From using two's compliment, it is given that the sum of a number x and the corresponding negative number y is equal to $2^N$. The number y should be converted into -x:

$$-x = y - 2^N \tag{4}$$

A scaling factor is used to convert the binary signals to the corresponding measured values. The number of quantization levels are $2^N$. The measuring interval on all four measurements are symmetric, $A \in [a_{peak}, -a_{peak}]$. The step length of a quantized segment, q, can be calculated by

$$q = \frac{a_{peak}}{2^{N-1}} \tag{5}$$

The step length q is also the scaling factor. The received sample in MATLAB is multiplied by q to receive the measured value.

### 3.5 Direct memory access (DMA)

Direct memory access is used to bypass the processor and save the data directly into a memory. This is useful when the CPU is needed to process other things often and can't be interrupted.

### 3.6 Inter IC-sound communication ($I^2S$)-communication

I2S Philips standard is a serial communication standard that sends two separate samples each clock-cycle. A Left-Right clock, also called a word clock, with the sample rate frequency is used.

The state of the Left-Right clock determines which signal is being sent. This is useful because the current can be sent through the left channel and the voltage through the right channel while only receiving the data through one wire. The I2S communication standard is common when working with audio.

### 3.7 Laser Sensor Head

The sensor head generates a laser spot on target to be measured. An high accuracy lens unit will focus the reflection of the laser spot onto LI-CCD sensor. A picture of the sensor head can be seen in figure 4.

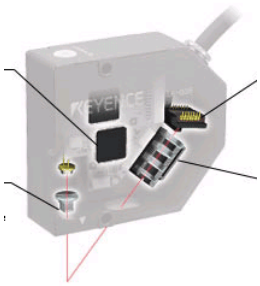

Figure 4: The laser diode generates a laser spot which is reflected onto the LI-CDD.

Depending on which pixel the light receive onto will result in different length measurements. The light in two pixels will generate a mean value of the two pixels for smother signal. A picture of how the light reflects into the sensor can be seen in figure 5.

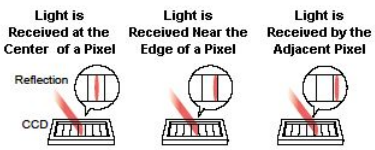


Figure 5: The reflected light in two different pixel and between the pixels. [6]

### 3.8 Circular buffer

A circular buffer is a fix size buffer which is connected in the start and end. Data will be written into to the buffer index ahead of the reading pointer. When the writing/reading pointer has come to the last index of the buffer it will goes to the starting index again. This can be seen in figure 6.

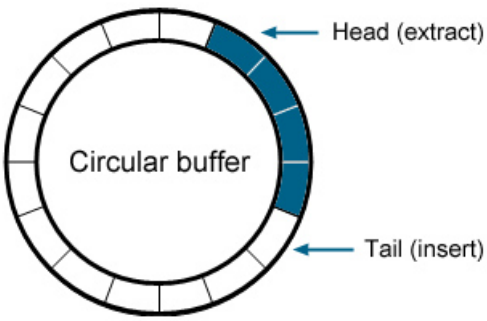


Figure 6: The writing pointer (tail) will fed data into the buffer and the reading pointer (head) will read the data from the buffer.

A circular buffer is useful when the processor is busy, but still needs to collect data for later processing. For example when writing on a keyboard the characters will be stored in a circular buffer before being used.

# 4 Method

## 4.1 The full system

An analog audio signal is fed into the amplifier and measures the current and voltage. These measurements are sent to the STM32. The current in the speaker will produce a sound signal and a microphone located close to the speaker will measure the sound pressure. The microphone is connected to a preamplifier which amplifies the signal. A sound card will receive the amplified sound signal and send it to the STM32. The STM32f4 will buffer all the measurements and send them to MATLAB for processing.

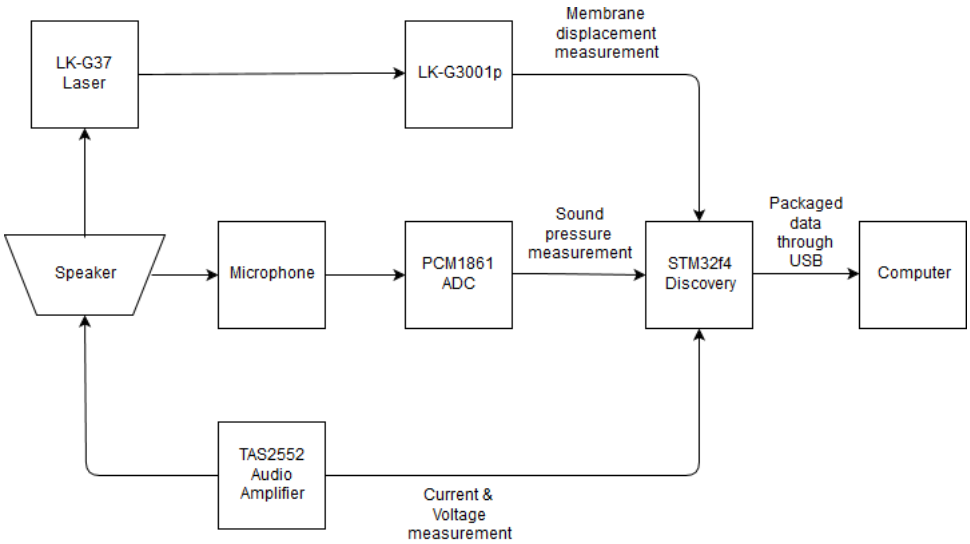


Figure 7: A block diagram of the final system, with all the measurements.

## 4.2 Speaker membrane displacement measurement

The LK-G30001P is used to measure the displacement distance. The data is transfered with an 40-Pin data cable to the STM32f407. Using the 21 data pins and an interrupt pin the data is collected and sent via USB to computer and then processed in MATLAB.

The Laser settings are configured with associated program LK-navigator. The laser is therefor required to be connected to a computer through USB. Without that connection, the stm32 and the laser will not have same ground which is needed. Through the associated software, a number of setting can be changed, like the sampling frequency, a number of filters can be applied and so forth. There is also a function where 64000 samples can be saved and plotted through USB. The data is saved as a .csv-file which also can be read in MATLAB.

The data is transmitted in two's compliment (from datasheet of the LK-G3000-series), which is converted to a positive or negative value in millimeters.

An interrupt routine in the laser will occur on rising edge when the strobe pin from the laser turns high. At the same time the laser measurement data will be available on the 21 data pins for 10 us. It is then very important that this data is read as soon as possible before new data arrives. From the beginning, the laser were placed in the same interrupt subgroup as the USB, therefore this two routines will stack on each other as a queue rather than interrupt each other. The laser interrupt was therefore placed in a separate subgroup with highest priority which made it possible to interrupt the USB.

## 4.3 Miniature speaker amplifier and I/V sense

Texas instruments provides an 4 W Class-D amplifier with built in I/V sense (TAS2552). An analog or digital signal is sent to the amplifier. This powered signal is sent to a miniature speaker which converts it to a sound signal. An associated windows software was used to configure the registers of the amplifier. By default the amplifier is configure to receive a digital input signal through USB. This was changed to receive an analog signal through the RCA input, but the user of the final setup can choose to use the digital input as well.

The I/V sense was sent to the STM32 through I2S, which the amplifier is configured as the slave and the STM32 as the master. Each frame of the I2S signal consist of either current or voltage measurement. Depending on whenever the LRCK is high or low either current or voltage will be sent. This is demonstrated in figure 8.

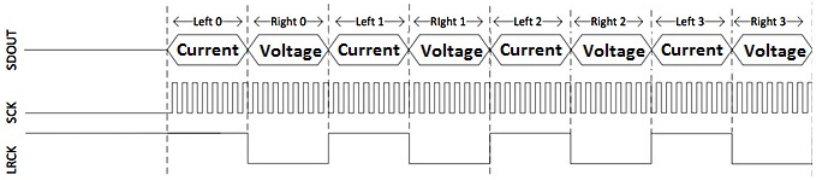


Figure 8: Depending on whenever the LRCK is high or low either current or voltage will be sent on the DOUT pin.

## 4.4 Sound measurement

A condenser microphone is used to record the sound pressure from the speaker. However the microphone needs 48V phantom-powering and a preamplifier. The microphone is therefore connected to a microphone amplifier which provides the phantom-powering and pre-amplifying. This amplified signal is then connected to the PCM1861 which is a hardware controlled audio ADC with 8 analog inputs and I2S output for the measurements. Each frame of the I2S signal consist of four 24-bit measurements. In this case, only one 16-bit measurement is needed. In the figure 9 the I2S protocol can be seen in for the PCM1861.

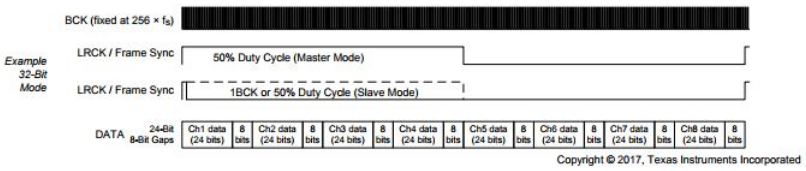


Figure 9: Each frame of the I2S signal consist of four 24-bit measurements [7].

The sound measurement is sent through I2S to the STM32f407 which is configured as the master and the PCM1861 as the slave. By using the hardware controlled interface for the PCM1861 it is configured in mode 2 which is demonstrated in figure 10.

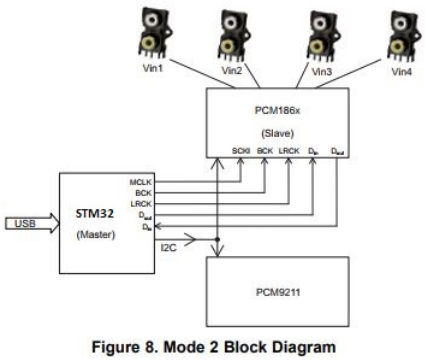


Figure 10: The PCM1861 is configured as the slave in mode 2, the PCM9211 is not used. [7] [10].

## 4.5 Receiving the I2S signal at STM32

The STM32 micro-controller provides two separate I2S channels and two associated DMA channels, one of these channels are connected to the amplifier and one to the sound measurement. To be certain that the sampling of the I2S occurs within the same sampling period, the I2S is configured in circular mode. In order to not interrupt the processor of the STM32 the I/V-data and sound-data is stored in a four sample DMA buffer each. If the DMA is half full (two samples) an interrupted will occur and two samples will be stored in a 16 sample circular buffer for each measurement. When the DMA has received all the four samples, a second interrupted will occur and the next two samples will be stored on the next index of the circular buffer. This way there will be plenty of time for the processor to do other things before it needs to empty the DMA.

Each time the laser-interrupt occur, one sample from each measurement will be collected from the corresponding 16 circular buffer. If the 16 samples buffer doesn't receive a new sample before the laser interrupt occurs, it will read from the same index again. If this happens an error byte will be set to a value corresponding to the missing measurement. More on this is explained in the chapter 'Data storage and conversion'.

## 4.6 Synchronization

The laser-measurement uses a 50kHz clock for sampling. The setup for this was already created and could not be changed. On the other hand, the I/V- and the sound pressure measurements, has a 48 kHz sample clock frequency. This creates a problem of synchronization and when the same micro controller collects the data, a choice have to be made of which clock to use to save every sample. If the 48 kHz clock is used, it will sometimes miss to collect a sample from the laser and data is lost. If the 50 kHz clock is used, the I/V- and the sound pressure measurements will sometimes be saved twice. Because no data should be lost, the latter method was used. The 50 kHz clock from the laser was chosen as the interrupt-trigger where every sample is saved in a buffer. As seen in segment 1, in figure 11, the 50 kHz clock gets triggered twice between one period of the 48 kHz clock. This would correspond to the micro controller saving the same sample twice into the buffer. A function have to be created to remove these extra samples somewhere in the process.

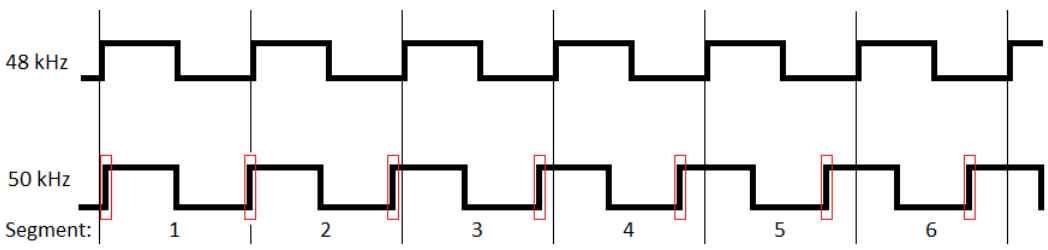


Figure 11: Two clock-signals, simulated as 48 kHz and 50 kHz frequency.

## 4.7 Data storage and conversion

The laser sends a 50 kHz sampling clock to the STM32 for each measurement that is done. This signal is used as an interrupt signal to store all data in a buffer. The sampling-rate of the I/V- and sound measurement is slower. Because of this, the interrupt will sometimes be triggered before new data has been saved in the DMA. This will lead to two saved samples of the same measurement in the buffer. To keep track of these double samples, an extra byte was added for each measurement. The error-byte gets its value through two global variables. They change value whenever a new measurement is saved in the DMA. By checking if these variables has changed value, the error-byte gets a specific value.

A start value of three bytes is saved first in the buffer with a value of FFFFFF in hexadecimal. Then a 21-bit laser-measurement-value is saved as 3 bytes. Two 16-bit I/V measurement-values are then saved as 4 bytes and a 16-bit sound measurement-value as 2 bytes. Finally, the error byte is saved. The package of data sent is 90 measurements of each, with an corresponding error byte and one start value. This makes the package 90*10+3=903 bytes. Figure 12 illustrates how the bytes in the package are inserted.

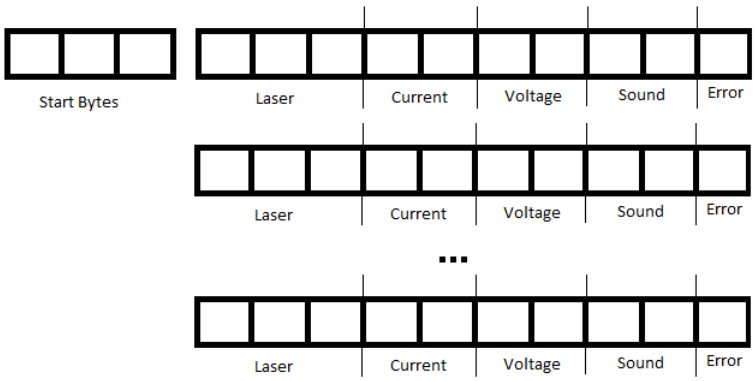


Figure 12: A diagram of how the bytes are inserted into the buffer in the STM32f4-discovery.

Packages of data is continuously sent from the micro-controller to the computer through USB. A python script run in MATLAB is used to collect the data. The data is read as ASCII, and converted to 8-bit unsigned integers. A time T defined by the user determines for how long the receiving period is. When the time is up, the separation and conversion of the data begins. To start the data conversion, the first start bytes has to be detected. This is done by searching for three bytes with values 255 in a row. When found, the program also checks if there is 900 bytes between the first and the next start value to keep the data synchronized.

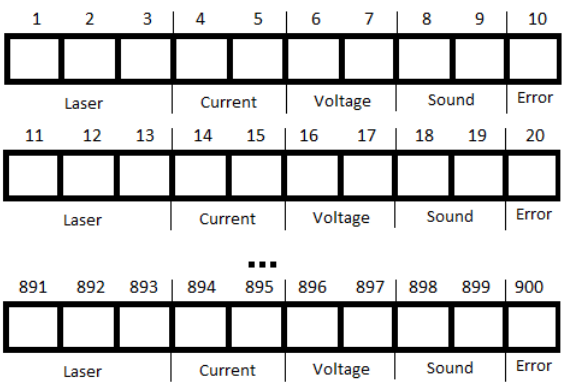


Figure 13: A diagram of how the indices are used to find the data from different measurements

A function is used to create an index-array corresponding to each measurement. For example,

from figure 13, the current index-array would be $\{3, 4, 13, 14, ..., 893, 894\}$. However, the error-byte decides if a current-measurement is valid or not. So if the error-byte has a value corresponding to the I/V-measurement not being updated, the index-array would not include the index values for these bytes. An example of this can be seen in figure 14. The index-arrays are then used to separate the laser-, current-, voltage and sound data in four separate arrays. The data is then converted from two- or three bytes to a signed whole number. The data is then scaled with $\frac{a_{peak}}{2^{N-1}}$, with $a_{peak}$ being the maximum value being measured (from data-sheet), and N is the resolution in bits.

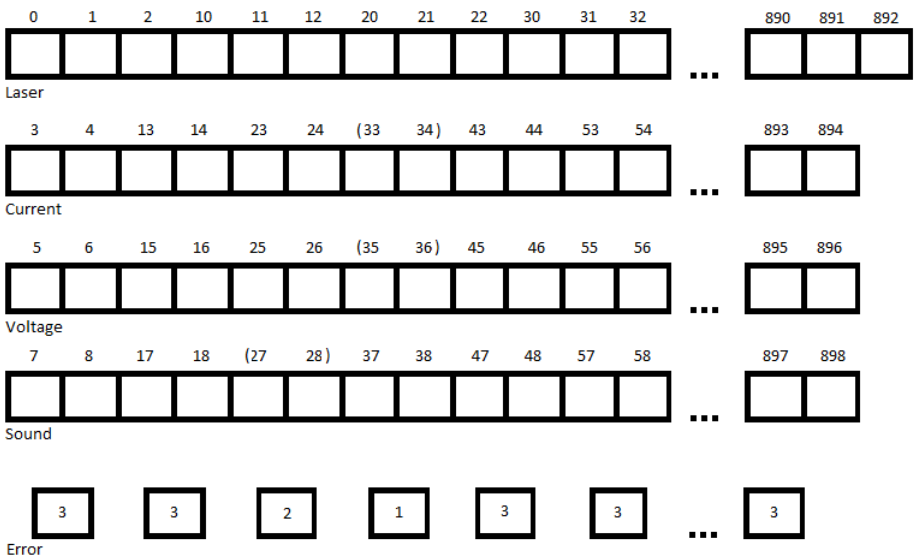


Figure 14: A diagram of how the error-bytes are used to remove certain samples. An error-byte with value 2 will remove the sound sample. An error-byte with value 1 will remove the I/V-samples.

# 5 Results

## 5.1 Laser

The signal from the laser to the micro controller is sent in parallel in a 40 pin flat cable. When measuring the pins receiving the signal, a problem was discovered. The signal had a very slow fall-time, which would lead to, at 50 kHz sampling frequency, the signal never reaching ground. The reason for this was found to be the length of the cable connecting the LK-G3001P and the STM32f4, and the internal pull-down resistors in the STM32f4 which were 40 kΩ. The capacitance in a 40 pin connector wire can be approximated by 50 pF/m. [11] The length of the wire was approximately 3 m long. This will create a lowpass-filter with cutoff-frequency of $f_c = \frac{1}{2\pi RC}$, with R = 40 kΩ and C = 3m*50pF/m= 150 pF. The cutoff-frequency is then around 26 kHz. The fall-time will behave like $v(t) = 3.3 * e^{\frac{-t}{RC}}$. The curve of the fall-time can be seen in figure 15. This fall-time corresponds well with what is seen in the oscilloscope, which can be seen in figure 16. The inductance in the wire was calculated to be to small to give any real effect on the signal.

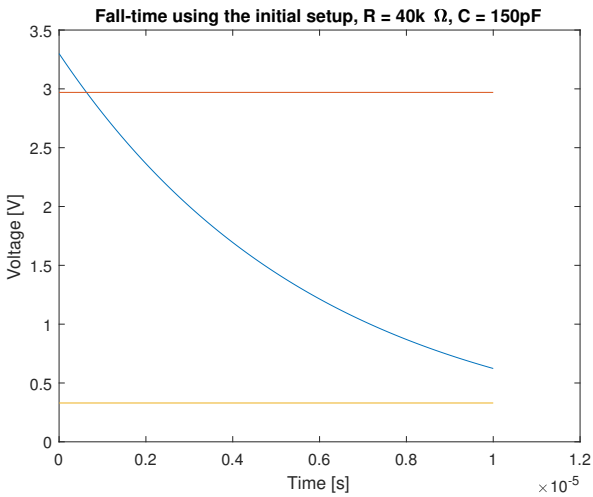


Figure 15: The fall-time shown in the time-interval of half a period of the interrupt signal from the laser. The cable capacitance is set to 150 pF and the pulldown-resistance is set to 40 kΩ.

Figure 16: The fall-time measured on the oscilloscope, with a long wire and the inbuilt pulldown resistance of 40 kΩ.

By using external pull-down resistors and by shortening the cable, the fall-time could be decreased as seen in figures 17 and 18. The theoretical graphs does not completely follow the measured decrease in fall-time but the results were improved. One problem was that the signal was sent in parallel through 22 cables from the laser. To get a good result, every cable would need a external resistor connected to ground. A resistor bank was soldered on to every pin connected to the laser.

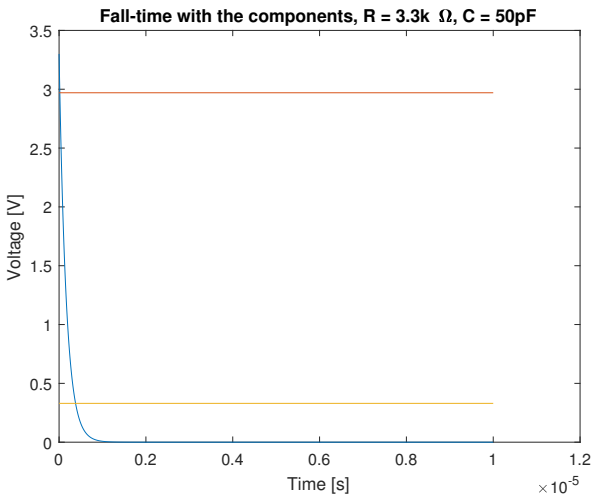


Figure 17: The fall-time shown with shortened cable and external resistors in the time-interval of half a period of the interrupt signal from the laser. The cable capacitance is set to 150 pF and the pulldown-resistance is set to 40 kΩ.

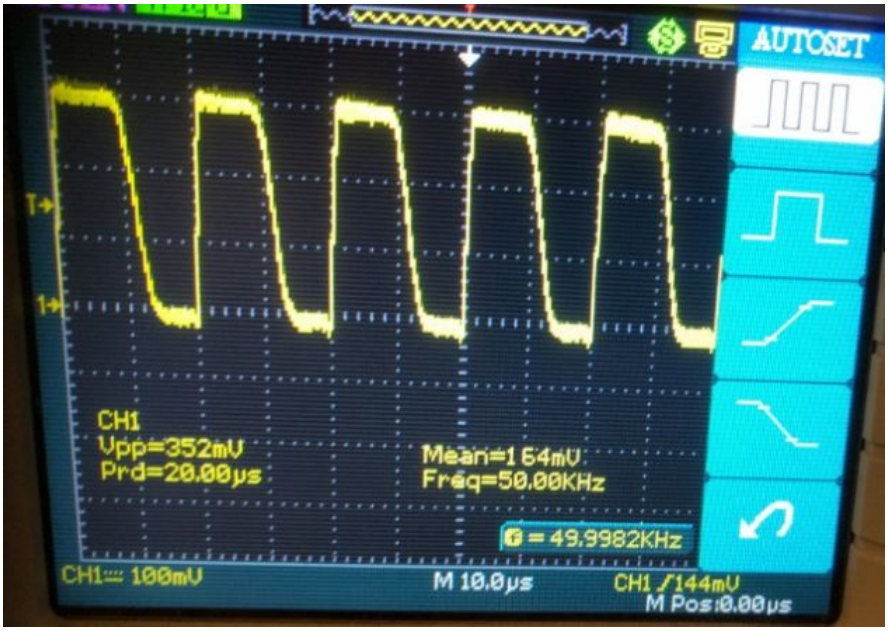


Figure 18: The fall-time measured on the oscilloscope, with a shortened wire and an external pulldown resistance of 3.3 kΩ.

Noise-sources was removed. For example, the power supply to the computers used made the signals very noisy and by removing them, the spikes became less frequent. The added noise on the trigger from the laser sometimes triggered the interrupt as seen in figures 19 and 20. This would lead to extra faulty samples being saved. The power-supply to the STM32f4 was initially through USB from the computer. This was changed to a new, independent power supply which somewhat stabilized the interrupt-signal.

Setting a new common ground between the stm32f4 and the LK-G3001P was tested, with no obvious difference. A different stm32f4-card was also tested, with no difference.

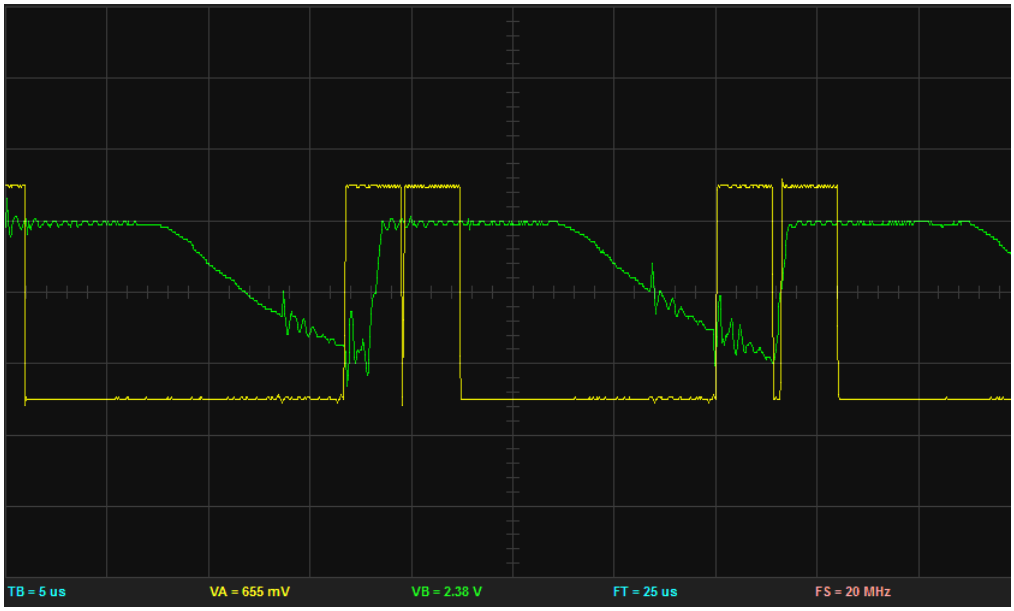


Figure 19: The green Laser-interrupt together with a yellow toggle when the data is saved. The power-supply to the computer is inserted, which lead to a very noisy signal

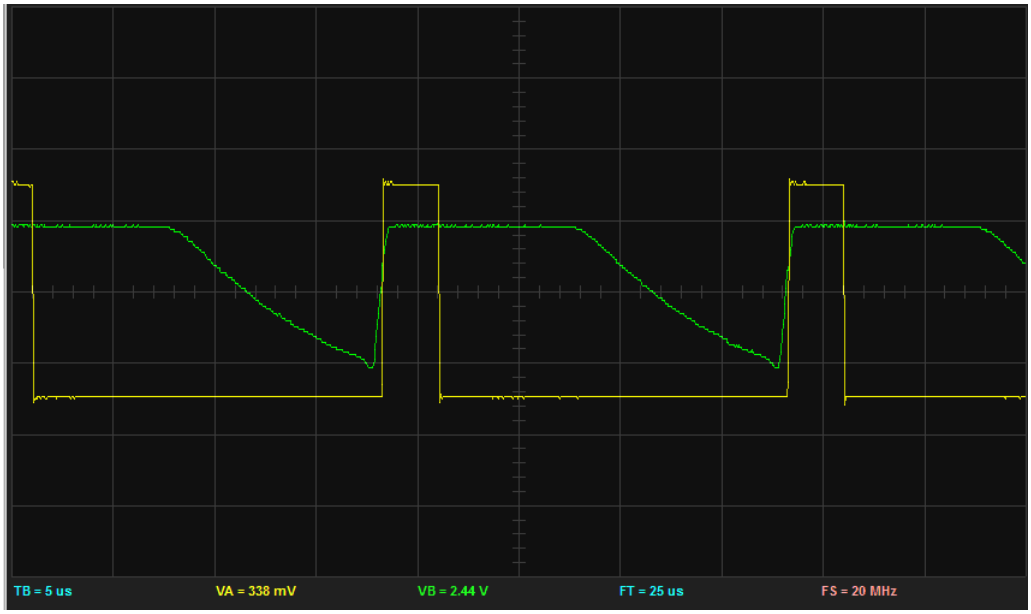


Figure 20: The green Laser-interrupt together with a yellow toggle when the data is saved. The power-supply to the computer is removed, which makes the signal less noisy.

A bit-map was created and showed no obvious problem in any single bit. However, it showed that whenever a spike occurred a large valued bit changed value. This would often lead to many small bits changing value as well. The bit-map also showed that in all cases, the value was higher than it should. This would mean that the pins would always be read as high when they in fact should be low. Some spikes seemed to come in a cluster where they all had an interval n*B samples, where n is a hole number and B is the number of laser-measurement samples in a sent package.

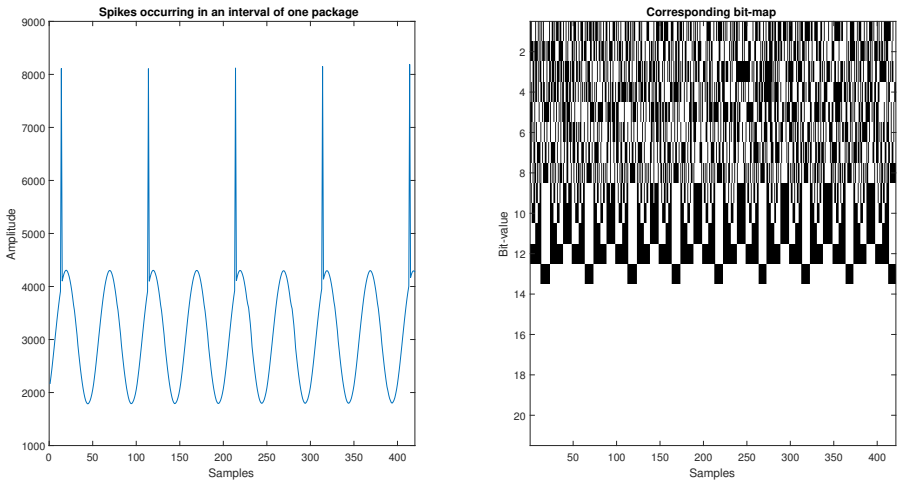


Figure 21: Part of a frequency sweep measurement, were a cluster of spikes occurred. As seen in the bit-map, no obvious bit is erroneous. In this case the number of samples in a package is 100.

From figure 21, all the spikes seemed to have the same amplitude. Zooming in on the spikes in the bit-map, the first spike seen in figure 22, a pattern emerges. The sample before has bit 13 as a zero, while the sample after has bit 13 as a one. When subtracting the value of the 13:th bit from the signal, the signal looks much better as seen in figure 23. In most cases while studying the bitmap, a large bit were erroneous, but not always. Sometimes several bits just below the large bit.

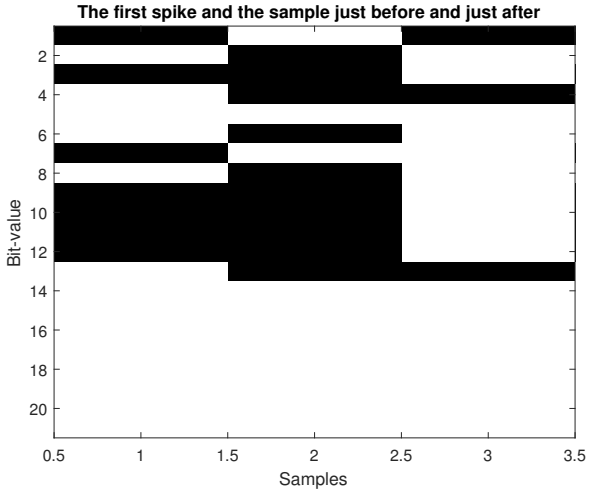


Figure 22: Zooming in on the bitmap where the first spike occurs.

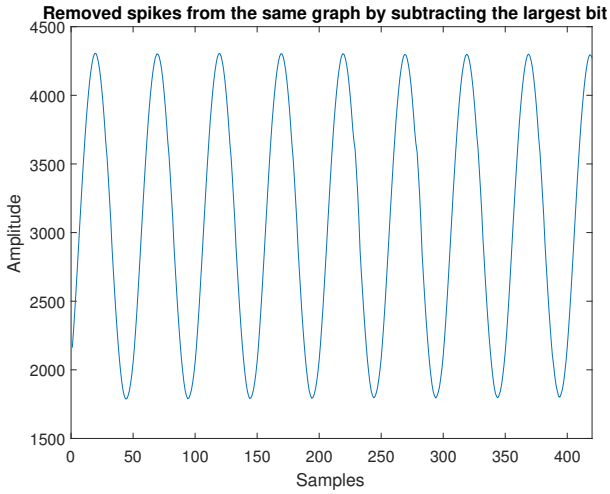


Figure 23: The signal with bit 13 removed from the samples with spikes.

Because the spikes sometimes occurred with the interval of a package-size, the time of the sending process was checked. The interrupts on the laser was toggled to check if the sending process was fast enough. The laser has a sampling rate of 50 ksamples/s which corresponds to 20 $\mu s$. The toggle took around $2\mu s$, which is around 10 percent of the time between measurements. This was deemed to be fast enough and not a problem.

By changing the position on the measuring plate beneath the laser during the measurement, spikes could from time to time be triggered. Also by creating a vibration in the laser, by for example slapping the table gently or stomping the ground spikes could sometimes be triggered. When removing the laser from the bar it is mounted on and laying it on the ground, spikes were removed and the noise was smaller. The bar the laser is mounted on works like a mechanical amplifier, and will start oscillating from very small disturbances. Because the laser is very sensitive, a small oscillation in the bar will increase the noise. Also, as seen in the bit-maps, whenever a large jump is measured from one sample to an other, there is a chance of a spike. A larger noise will make this happen more often. That is why, at higher frequencies, the spikes were more common.

Even thought the spikes was sometimes in an interval, they often seemed pretty random when they occurred. These spikes was harder to find a pattern in but they all had in common that they happened when one large bit changed value. An oscilloscope was used to check the interrupt-time whenever a large spike occurred. The external trigger on the oscilloscope was used to save data

whenever a value on a noise-signal got a value larger than 100 from the previous value. From figure 24, it is clear that the interrupt is delayed by a large fraction of the period-time. This would lead to spikes.

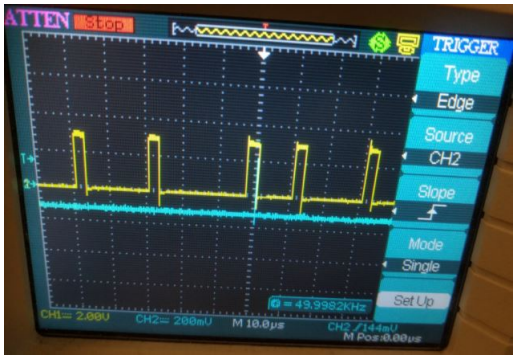


Figure 24: The yellow graph from the oscilloscope is whenever the laser-interrupt is triggered. The blue graph declares a value that is unusually high, in other words, it indicates a spike.

From the oscilloscope the interrupt routine from the USB was measured to 12 us. According to the datasheet of LK3001 the digital data from the laser is available for 10 us after an interrupt from the laser occurs. This will result in a conflict between the laser interrupt routine and USB routine. For example, if an USB interrupt occurs 1 or 2 us before the laser strobe happens, the CPU will be occupied with the USB routine for 12 us before it has time to read the digital pins from the laser. This will result in a mixture between the current laser sample and the next sample. This issue was solve by placing the USB and the laser in separate interrupt routine groups and set the priority higher on the laser interrupt. In this way the laser will interrupt the USB routine and store the laser sample before CPU returns to the USB routine again. The spikes was removed by implementing this.

The standard deviation was calculated in MATLAB for when the laser was mounted on the bar over 300000 samples, with and without the power supply to the computer inserted. A measurement was done With charger in, mounted on bar, no talking, very gentile mouse-click:

$$\sigma_{mm} = 5.8408 * 10^{-4}$$

Without charger, mounted on bar:

$$\sigma_{mm} = 5.5337 * 10^{-4}$$

The standard deviation was also calculated when the laser was stabilized and with no power supply inserted:

$$\sigma_{mm} = 9.6005 * 10^{-5}$$

The distribution of the noise was checked with a histogram. From figure 25, the noise has a Gaussian distribution.

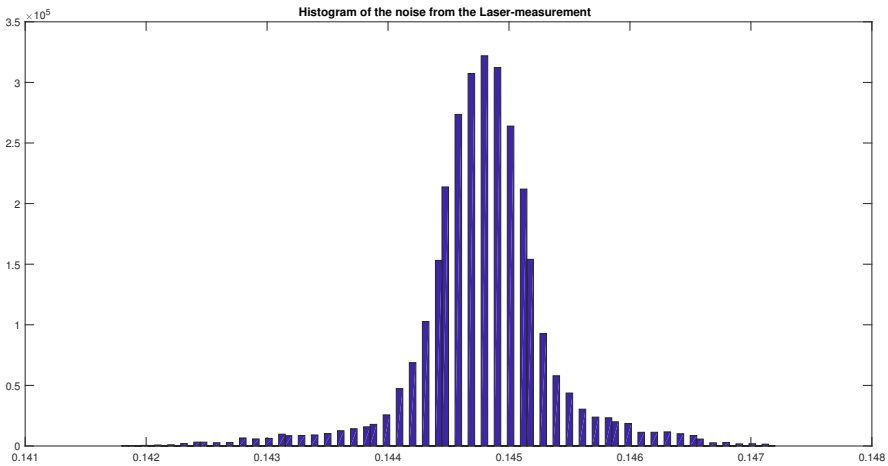


Figure 25: The histogram of the distribution of the noise from the membrane displacement measurement, when the laser was mounted on the bar, using over 3 million samples.

The spectral estimate of a frequency sweep from 80 Hz to 20 kHz was calculated, using a periodogram. This can be seen in figure 26

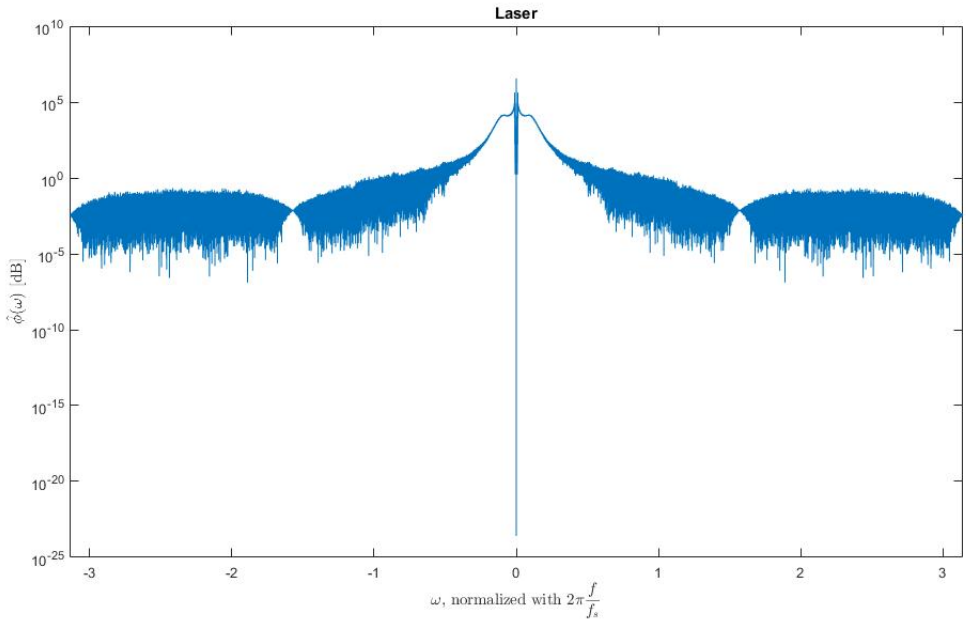


Figure 26: Spectral estimate of the speaker from a frequency sweep.

## 5.2 Current and voltage measurement

At first, to see if the amplifier worked, the measurements was read through a while-loop without using the DMA. After that, a DMA was set up and read from. A timer-interrupt was now needed to simulate the laser interrupt. The timer-interrupt was created to have a frequency just over 50 kHz. The signals measured now had some extra samples with the same value, as expected. However to many of these double samples occurred. By doubling the frame-window of the Left-right-clock to 32-bits, this problem was fixed.

The peak current and voltage from the TAS2552 data-sheet is, $I_{peak} = 1.4A$ and $V_{peak} = 8.5V$. The resolution on the current and voltage is from equation 5,

$$I_{res} = \frac{I_{peak}}{2^{15}} = 4.27 * 10^{-5} A \tag{6}$$

$$V_{res} = \frac{V_{peak}}{2^{15}} = 2.59 * 10^{-4} V \tag{7}$$

A sweep from 80 Hz to 20 kHz was measured, with close to maximum power. The signal measured is plotted in figure 27. In figure 28 the signal is zoomed in to about two periods. The sinusoid looks very clean, and can easily be calculated to be around 400 Hz at that point.

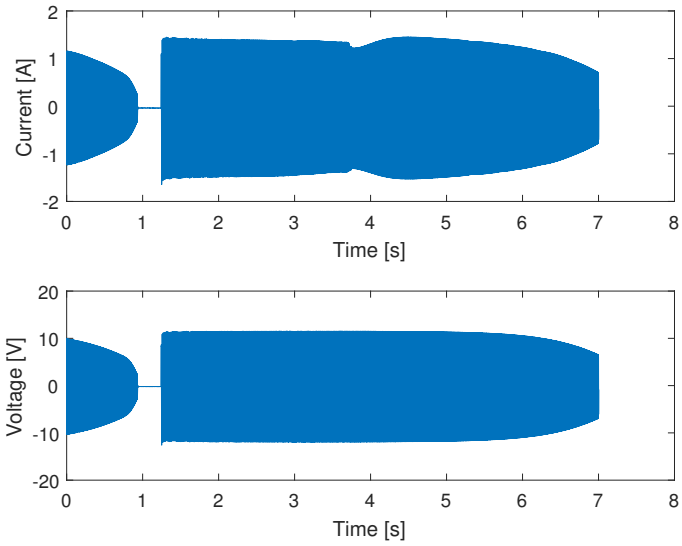


Figure 27: A measurement of a frequency sweep signal into the audio input.

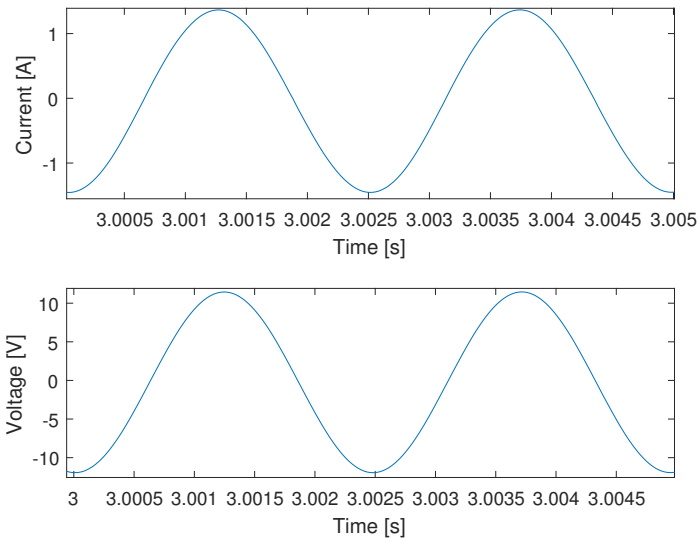


Figure 28: The measured signal zoomed in after three seconds.

The spectral estimate of the frequency sweep is done by using a periodogram with a rectangular window function. The spectra can be seen in figure 29 and 30.

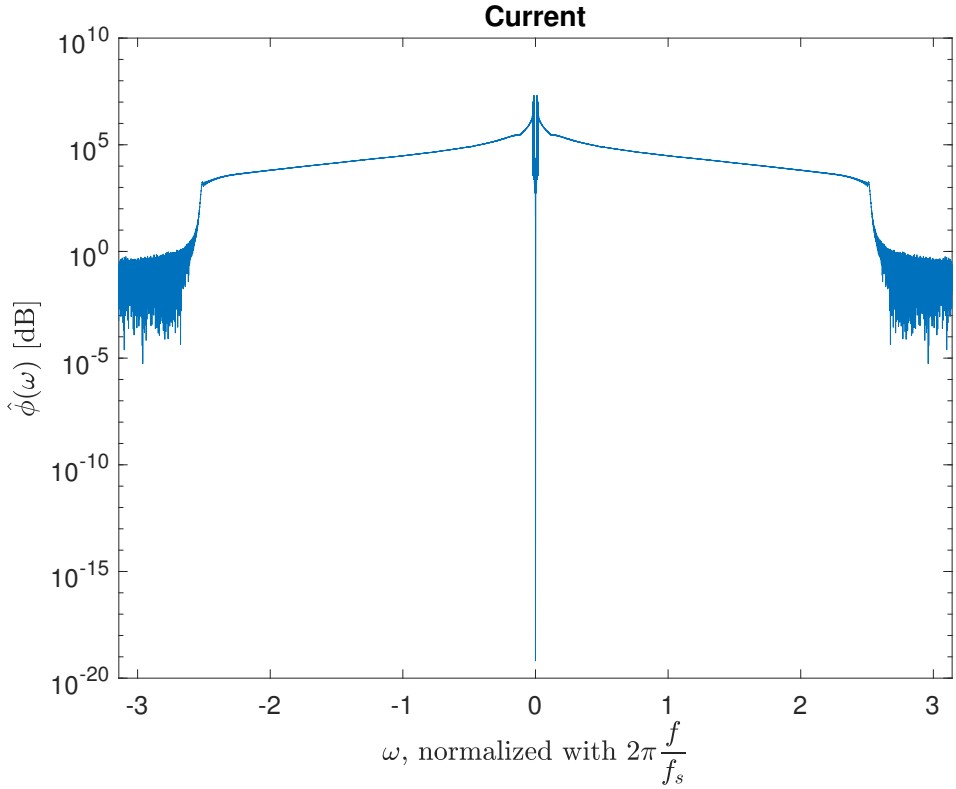


Figure 29: Spectra of the current from a frequency sweep.

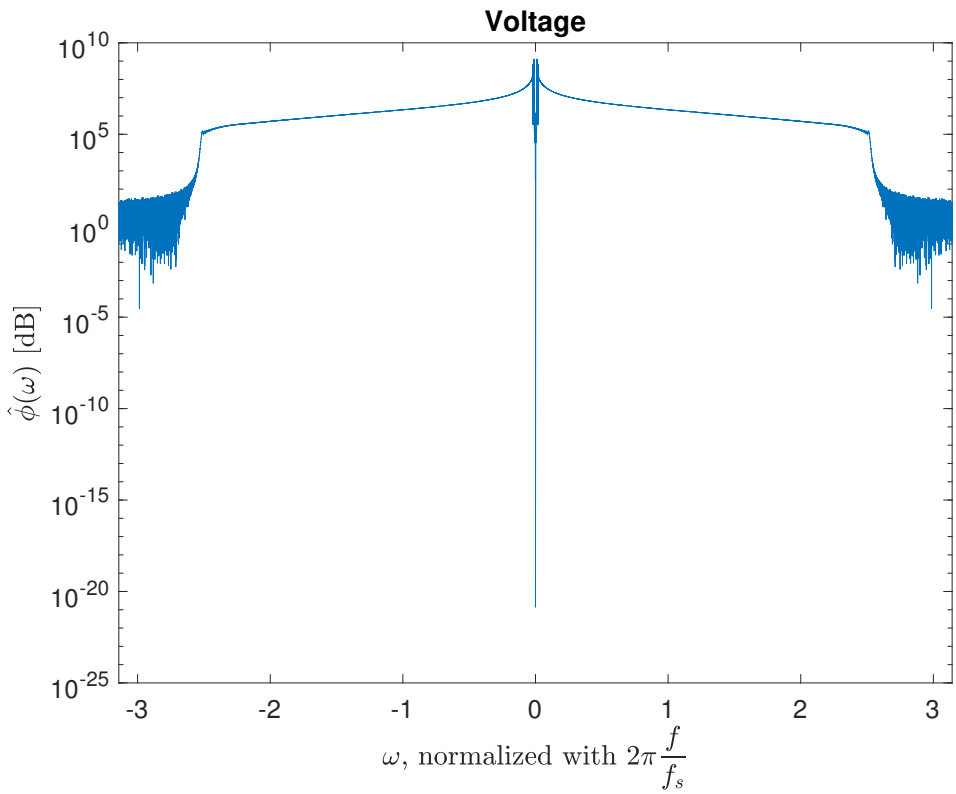


Figure 30: Spectra of the voltage from a frequency sweep.

The noise of the current and the voltage was measured, seen in figure **??**. The standard deviation

was calculated and a histogram of the noise was plotted seen in figures 32 and 33. The current-noise standard deviation is around 6 times the quantization level The voltage-noise has 8 times the quantization level.

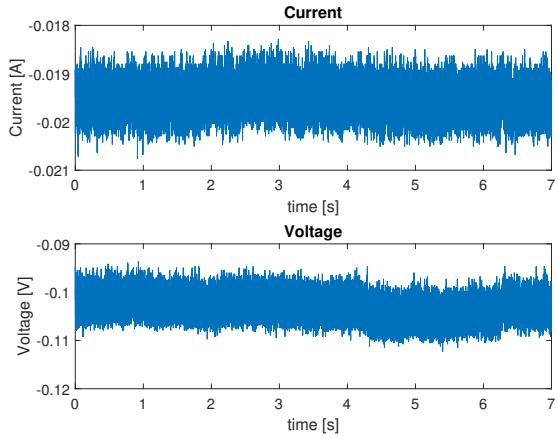


Figure 31: The measured noise from the current and the voltage.

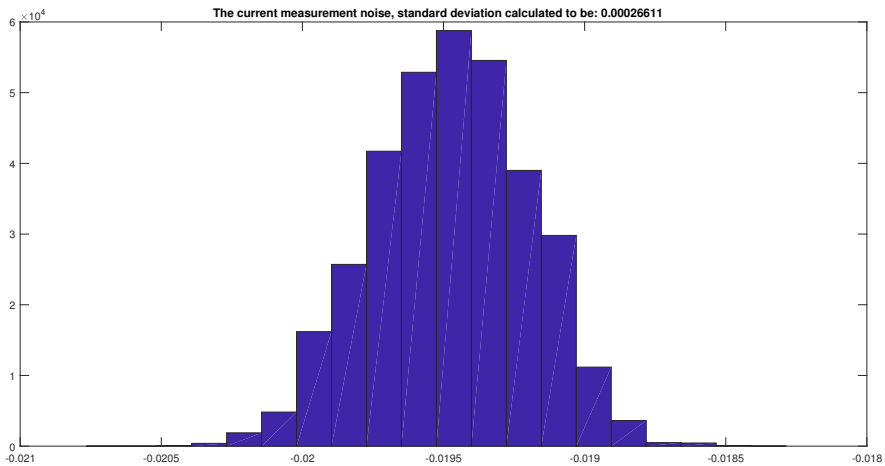


Figure 32: A histogram of the current measurement when no signal is inserted into the speaker.

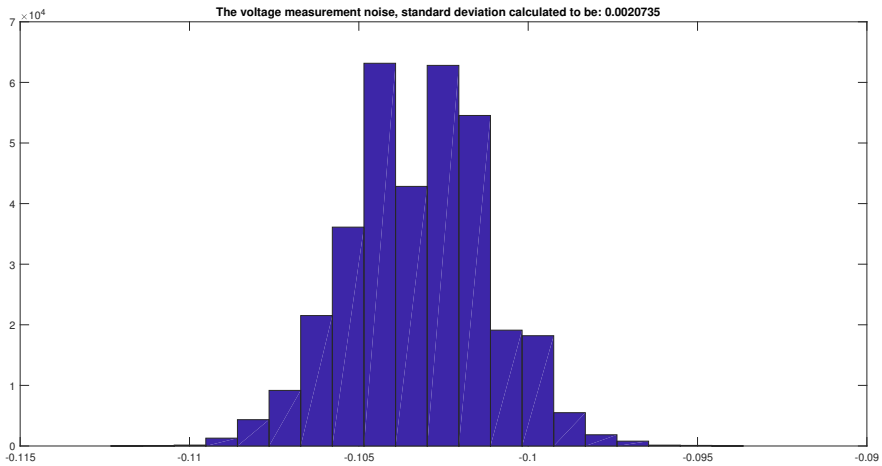


Figure 33: A histogram of the voltage measurement when no signal is inserted into the speaker.

By using the measurement of the frequency sweep and taking the peak values of the voltage divided by the peak values of the current at different frequencies, the impedance was measured as seen in figure 34. The impedance increases at higher frequencies because of the coil, with a resonance peak at around 800 Hz.

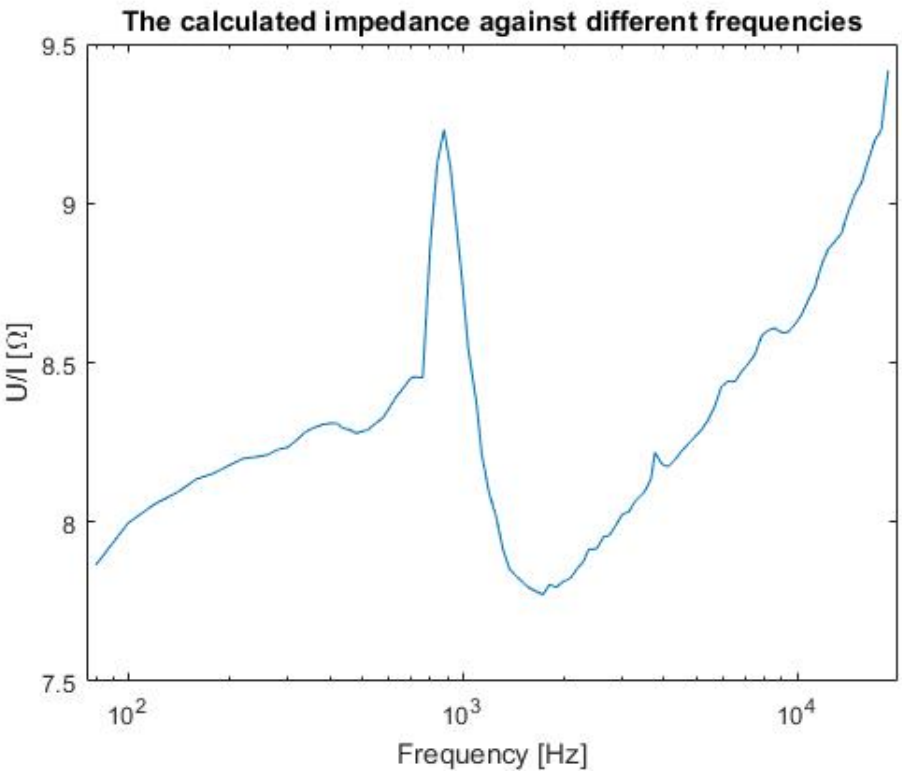


Figure 34: The impedance at different frequencies.

## 5.3 Sound pressure measurement

The sound pressure measurement was successfully implemented using The PCM1861 and I2S. The sound pressure was sent with an sampling rate of 48.8315 kHz and stored into the DMA buffer. The data was then emptied into the circular buffer and sent to the computer via usb. A sine-wave sweep was fed into the speaker from 70 to 20kHz and measured with the microphone and plotted

in MATLAB. This can be seen in figure 35.

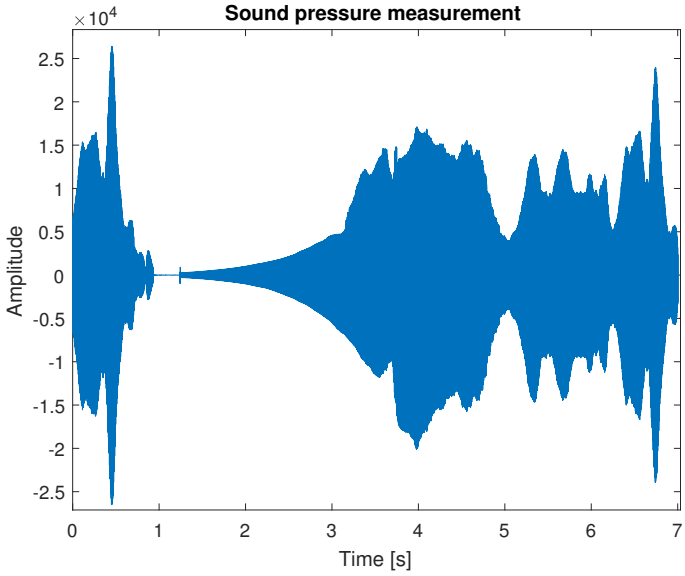


Figure 35: Sound measured from 70 to 20 kHz.

A closer look of 400 Hz from the sound pressure sweep can be seen in figure 36. However the signal seem to be a mixture of several sound signals which gives an faulty sinusoid waveform. This happens because the sound is measured in a 90 degree angle close to the speaker which will cause the microphone to receive reflections of the sound.

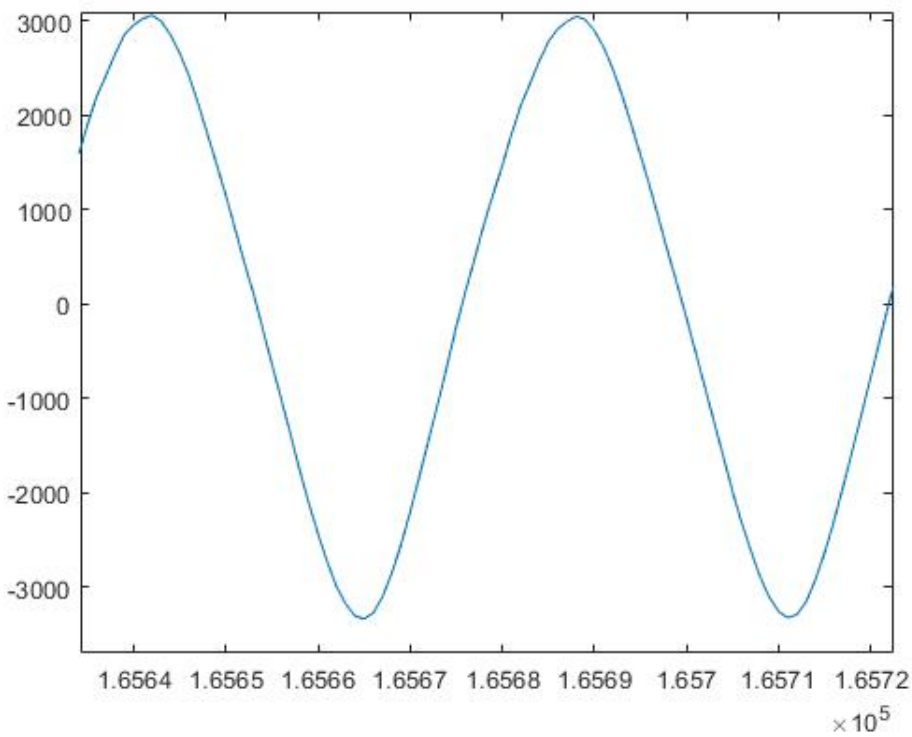


Figure 36: Sound measured of 400 Hz sinusoid.

The sampling clock to both the sound- and the I/V-card was measured seen in figure 37. The blue curve is the sampling clock to the sound measurement. It was measured to be 48.8315 kHz and

very little jitter according to the oscilloscope used. The same clock was used for I/V-measurements with the same frequency. Very little jitter was detected.

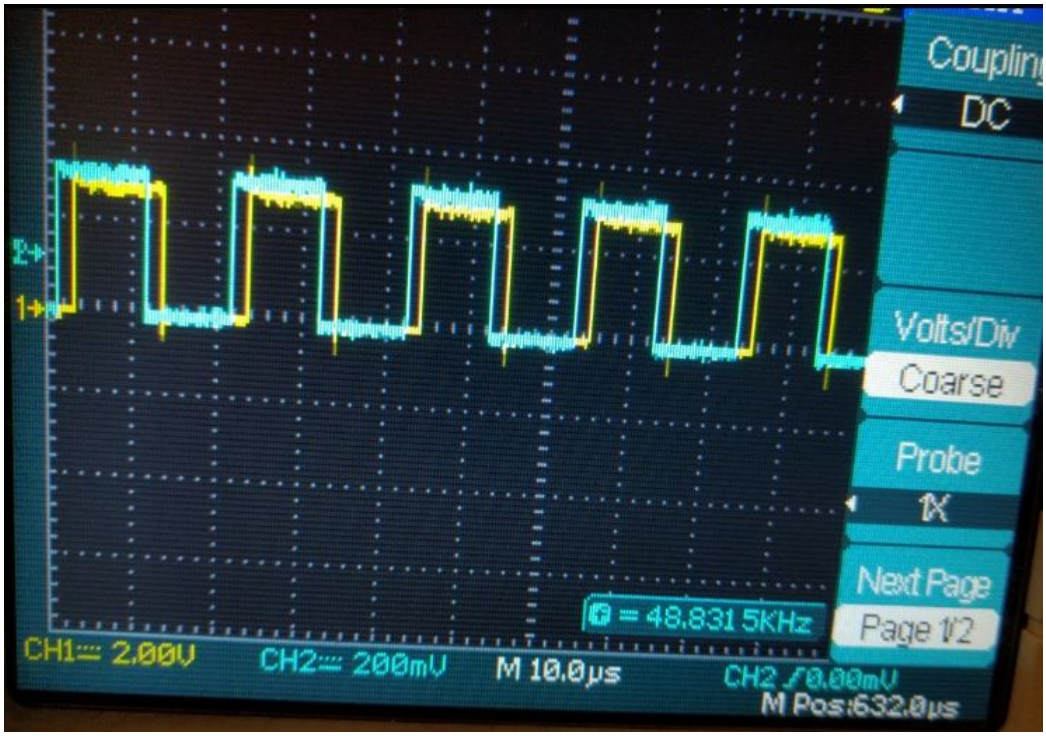


Figure 37: The Left-Right-clock signal from both the I/V- (Yellow) and the sound measurements (Blue).

## 5.4 The final setup

The initial setup of the laser-measurement, a 40 pin connector was connected from the LK-G3001p to the STM32f4. The I/V- and the sound-measurement needs 4 pins each, the Left-right clock, the bit clock, a master clock and a receive digital data from the I2S-communication. Only some specific pins can be used for this and some of them were initially used by the laser. A new setup was created where the pins required for the other measurements was freed.

The implementation of all the measurements at the same time created data-losses. To test the data-transfer speed, dummy-data was sent from the micro-controller in form of a saw-tooth wave. This was created by creating a counter that added 10 to the old value each time data was saved. Different package-sizes was tested to see what was optimal.

The required baud-rate was calculated. 10 bytes is saved every time data is saved into the buffer. The laser interrupt triggers the micro-controller to save 10 bytes at a frequency of 50 kHz into the buffer. This leads to an approximate baud-rate of 10*50000 = 0.5 Mbytes/s or 4 Mbits/s. This was a problem when using the MATLAB serial function. A python-script was written, which could receive the data without any losses. The MATLAB program then used the python-script to receive data in either a long array or written into a .txt file. The data transfer was toggled and showed that the speed was fast enough. The STM32 had no problem sending that fast and the Python serial function had no problem receiving data that quickly.

The maximum buffer-size in the STM32f4 was calculated. From STM32f4 data-sheet, the FIFO-buffer in the STM32f4 is 1.25 kBytes. The package-size that need to be stored there is calculated by #startbytes+#total bytes per sample * #of samples/package. The number of bytes needed from every measurement is, 3 bytes from the laser-measurement, 4 bytes from the I/V-measurement, 2 bytes from the sound pressure-measurement and one error byte. This adds up to a total of 10 bytes/measurement. Three start-bytes was used. This means the number of samples/package cannot be higher than 124. From testing with different buffer-sizes, the number of samples/package

used was found to be around 90 for a lossless data transfer, which would lead to a buffer-size of 903 Bytes.

The STM32 was fast enough to transfer the buffer into the USB-OTG buffer before the next sampling of the laser occurs which is 20 ms. This was validated by toggle a pin when the STM32f4 is transferring data into the USB-OTG buffer. In the picture below the broader square wave is when the buffer is transfered to the USB-OTG buffer.

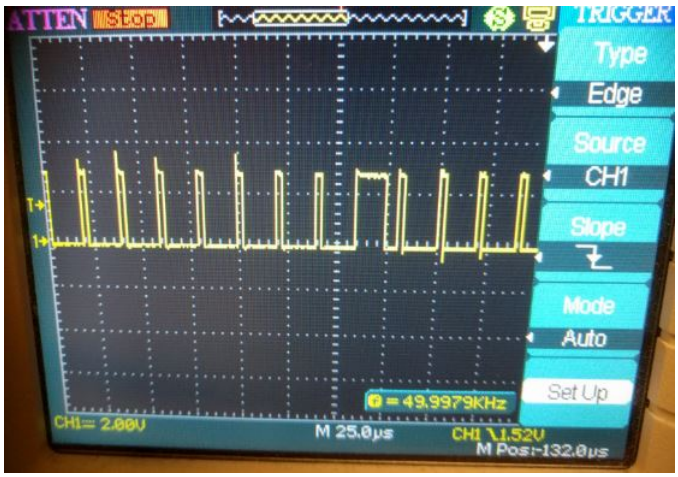


Figure 38: The broader square wave is when the buffer is emptied into the USB-buffer.

It took around 1 ms to send the data. When using a buffer of 120 samples, the number of packages sent each second should be the sampling frequency /samples/package: 50000/120=416.66 packages/s. From looking at the toggles a package was sent every 2.4th ms which corresponds to a package frequency of 416 packages/s. The time used to send a package is around 40% of the time-period.

The FIFO-USB buffer is 16 kBytes, from which around 12.6 kBytes was measured to be saved before the buffer was full in the python script. With a baud-rate of 0.5Mbytes/s the buffer would be full within 12.6k/0.5M = 25.2 ms. If implementing the real-time plotting of the data, all operations on the data must be finished before the buffer is filled. This was proven to be difficult as whatever was tried, large chunks of data was lost. A way to work around this would be to use parallel processing. The computer would receives data and process the data in two parallel processes. In MATLAB this could only be implemented by buying an toolbox which cost around 10000 kr. Dirac was not prepared to spend that money, so the implementation of real-time plotting of the data was put on hold.

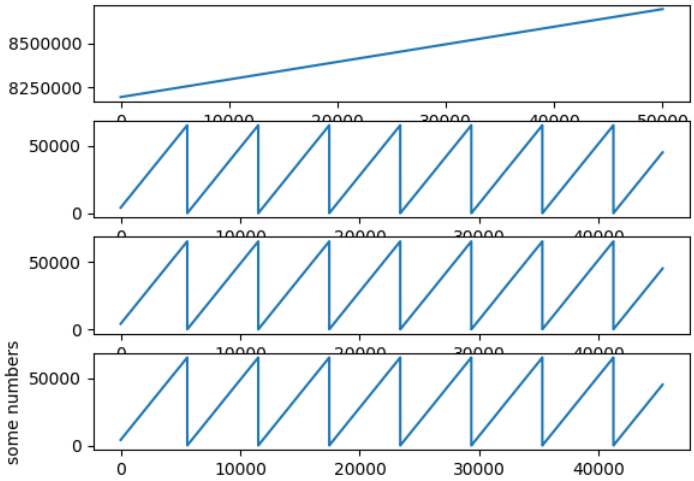


Figure 39: Dummy-data created in the micro-controller sent to Python. No data-losses anymore.

The laser-spikes started to appear again when everything was connected. For some reason, the I/V-measurement card disturbed the laser interrupt signal. From having a pretty constant frequency of (49.9982 kHz), the signal started to jitter with a frequency from 51-55 kHz. The same type of jitter was created when the power-supply was inserted into the computer. To stabilize the signal, an external power-supply was inserted into the micro-controller. The problem still remained, but was later solved, as seen in the laser section 5.1. A picture of all the measurements of a frequency sweep can be seen below.

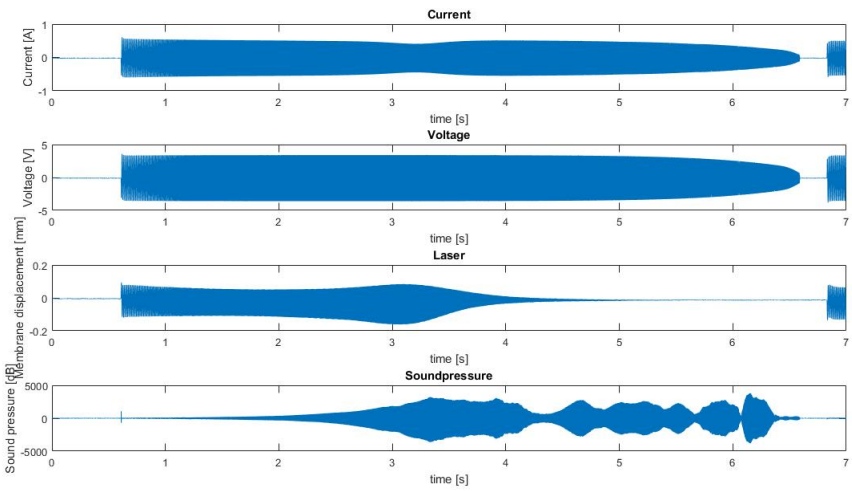


Figure 40: A graph of the four measurements done simultaneously. The spikes has been removed from the laser- and the extra samples has been removed from the I/V- and sound-measurements

## 5.5 Non-linear effects

The power of the signal was raised to the limit of the audio amplifier of around 4 W. This way, signs of non-linear behavior should be detected in the membrane displacement measurements. The signals was compared to simulated sinus-waves with comparable amplitude. Three measurements was done with a frequency of 20 Hz, with different power. A spectrum was estimated of a low frequency sinusoid measured by the laser to find some non-linear effects.

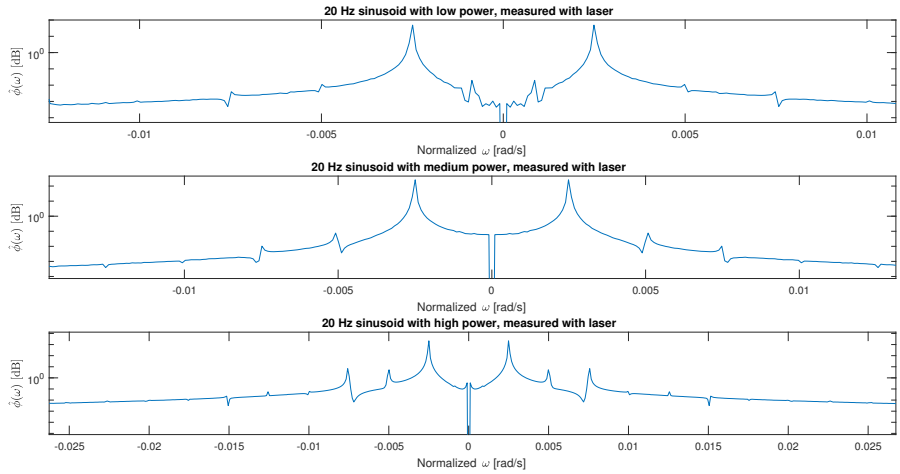


Figure 41: The spectral estimate of a 20 Hz sinusoid sent through the speaker and measured by the laser. The current was measured to be 1.3 A

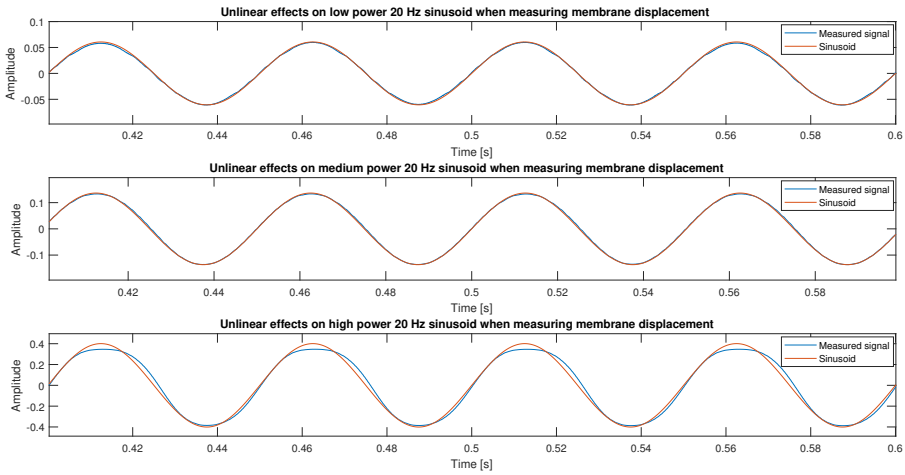


Figure 42: The measured signals plotted against a 20 hz sinusoid with similar amplitudes

From figure 41, the harmonics when using a low power wave was very small. By increasing the power of the signal, the 2nd and third harmonic increases when the power is high, and can barely be seen when the power is low. In figure 42, the non-linearity in the signal looks pretty evident. The high power signal is squashed at the top and asymmetric along the y-axis. When the membrane is far from its equilibrium point, several non-linear effects can have an effect. The stiffness constant K in Hook's Law, seen in formula 2, is dependent on the position of the membrane. When the displacement from the equilibrium becomes big, the stiffness factor K will increase. This will make it harder for the coil to move away from the equilibrium and the sinusoid will flatten. Likewise, the force-factor will decrease if the coil moves far away from the equilibrium point. This will also contribute to the flattened sinusoid.

The non-symmetric part of the signal can be explained by the varying inductance dependent on if the coil lies above or below the magnetic gap. A frequency sweep from 80 Hz to 20 kHz was measured over 6 seconds. The signal showed a clear picture of the anti-symmetric behavior, as seen in figure 43.

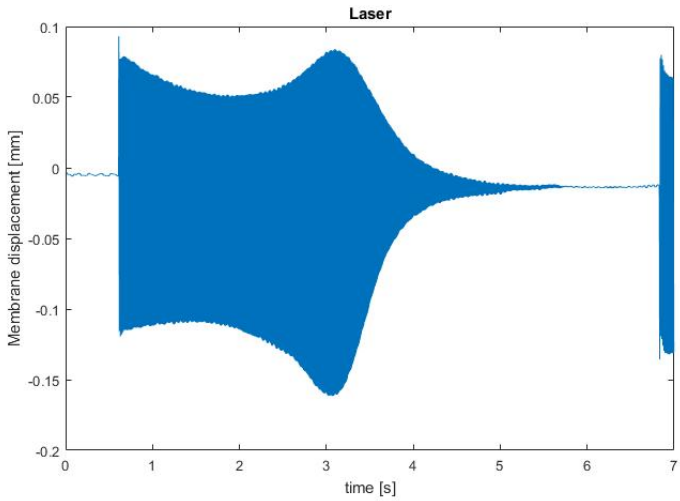


Figure 43: A frequency sweep from 80 to 20kHz through the speaker, measured with the laser.

# 6 Discussion

A different method of measuring the current and voltage was discussed. An analog amplifier, amplifies the input signal to the speaker. The current measurement is done by placing an shunt resistor in series with the amplifier and measure the voltage over the resistance. The voltage is measured using an ADC built for audio applications. Using Ohms law the input current through the speaker is calculated. The difficulty to measure the current is to not disturb the signal when the load is low-impedance. By using the solution above, these effects are minimal. This solution was dismissed because the final setup Dirac wanted was more in line with the method used and they were willing to pay for a more expensive solution with better results.

Initially, while implementing the DMA, 1-2 missing samples occurred right after the buffer was sent to computer through USB. By using the halfcpltcallback interrupt, to empty half of the DMA and then use cpltcallback to empty the final part. Using this function prevents data from being overwritten in the DMA-buffer.

## Removing multiple samples because of the clock speeds

Several ideas was contemplated of how to remove the extra samples. One idea was to remove the data in the STM32. A problem occurs when the data should be separated. The data sent must be structured so it can be separated easily. By removing some samples, it would be tricky to now which bytes belongs to which measurement. Some kind of information of how many bytes of each measurement must be sent.

Another idea was to change the value of every extra sample to some known value, possibly all 1s. A problem with this method is when the starting-bit is already a bunch of ones, the double samples could be confused for the starting values and data would be lost.
The third method, which finally was used, is to send one extra byte for each measurement. Instead of storing 9 bytes each measurement, 3 laser, 4 I/V, 2 sound pressure, you would add one error-byte which will be used in the data conversion. For a specific value of the error-byte, the measurement would either be saved or removed from the converted data. A problem with this is that the baud-rate increases by 10% and the buffer fills up faster. Since the buffer size from the USB-FIFO in window was shown to be a problem, the increase would not be optimal especially when trying to implement a real-time plot. Still, this method was used because the real-time implementation didn't work even with a lesser data speed and without real-time plotting no data was lost.

A problem occurred when implementing the algorithm of removing erroneous samples. The DMA sometimes saved samples twice, before the first samples was stored in the package-buffer. This would change the value of the error byte. The error byte would then tell the algorithm to remove a measured sample. This was detected when the converted data had jumps. The data also had different amount of samples from the sound- and the I/V-measurement, which is strange when using the same clock for measuring. A number of theories of why this happened was tested. The DMA-interrupts was tested so they weren't writing over the error-check value while the data was saved, which would lead to a correct measured sample being removed. The problem was solved by using a circular buffer with 16 samples. The two counters was used to save the data. These counters were used as global variables and the by comparing the value of the counters to the previous value inside the laser-interrupt, the error-byte was asserted a certain value.

## Synchronization

Some effort was spent on synchronizing the data by doing a re-sampling the laser at 48.8315 kHz. The function resample() was used in MATLAB, but could only approximate the sampling frequency by creating a fraction close to 50000/48831.5. This method re-sampled the Laser-measurements with a frequency close to the I/V- and sound measurements. A method would have to be created to line each corresponding sample for the four measurements.

# 7 Conclusions

One problem with the noise on the measurement value was remove by building a more robust setup for the laser. The only noise that could be observed after the laser was implemented in the more robust system had standard deviation of $\sigma_{mm} = 9.6005 * 10^{-5}$.

One of the most time-consuming parts of the project was trying to implement real time plotting. It took a couple of weeks of trial and error to realize the data input to from MATLABs serial function was not fast enough, which was shown as package-losses in the plots. When trying a python script instead, the data could be transferred fast enough, but not converted in real time without data-losses. The idea of parallel processing was born, but never implemented because of time and money as the add-on required in MATLAB was expensive. Parallel processing could possibly be done in python but not much time was spent on that.

The laser measurement took a lot longer than was to be expected. One problem was that we thought the laser spikes was solved early in the process, when only the measurements on the laser was done. The spikes seemed to disappear when removing the laser from the beam it was hanging from, and when the power-source to the computer was removed. Several measurements was done, with over 30 seconds of data each, with no spikes occurring. The problem probably was that the amplitude of the measurement was really small. As seen in the bit-diagrams, the spikes most often occurs when a large bit changes value. If the measurements oscillate around the smallest bits, no spikes would be created. Therefor, when doing real measurements, at higher power, the spikes started to show up again. This was finally solved by placing the interrupt routine from the laser in a separate sub priority group. However much of the final part of the project was spent to solve this problem.

Most of the time spent on this project was to create and fix the embedded-system part. Therefor, too little time was spent on actually analyzing and synchronizing the data. Even less time was spent on implementing the real time plotting, which was a shame.

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